

Week 6 — Activity: Magnetic Structures

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Purpose

This activity mirrors previous gravity workshops. You will reason about magnetic anomaly shapes and compare them directly with gravity anomalies from the same structures.

Assume:

- Uniform susceptibility
- Induced magnetisation only
- Constant inducing field

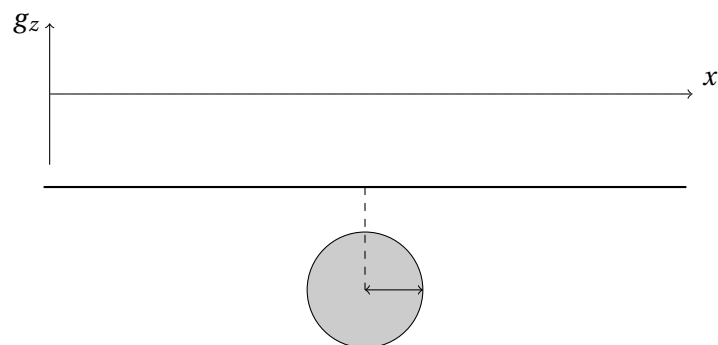
Instructions

For each structure:

1. Sketch the expected vertical magnetic anomaly
2. Indicate sign, symmetry, width, and strongest gradients
3. State how it differs from the gravity anomaly

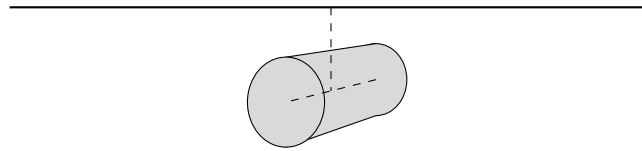
Structures

A) Buried Sphere (3D)

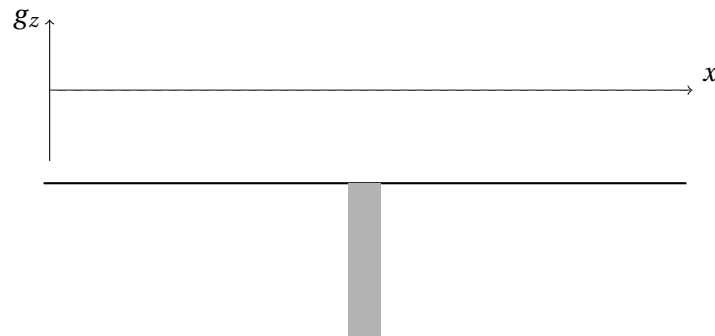


B) Buried Cylinder (2D; infinite strike)

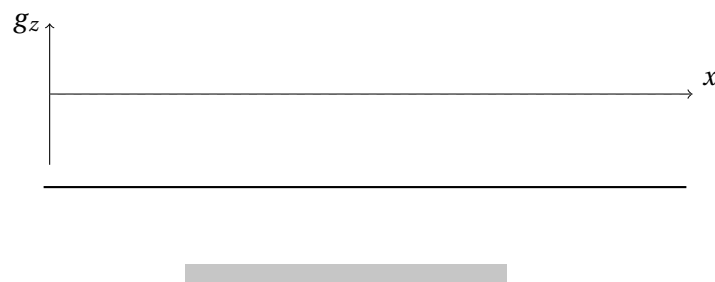




C) Vertical Slab



D) Finite-Extent Horizontal Slab



Vertical vs Horizontal Inducing Field

Repeat one case assuming a horizontal inducing field.

1. How does the anomaly shape change?
2. Which case appears more gravity-like?
3. Why does gravity not behave this way?

Reflection

Give two reasons why magnetic interpretation is more non-unique than gravity.